

GIRRAWEE HIGH SCHOOL
MATHEMATICS EXTENSION 2 TASK 1

December 2008

Instructions:

*Write all solutions on your own paper.

*Show all necessary working.

*Marks may be deducted for careless or badly arranged work.

*Approved scientific calculators may be used.

Time allowed: 90 minutes.

MARKS

Question 1 7 Marks

Given $z = 1 - 4i$ and $w = -2 + 5i$ find in Cartesian ($x + iy$) form:

- | | |
|---------------------|---|
| (a) $z + w$ | 1 |
| (b) $\bar{z}$ | 1 |
| (c) $\frac{z}{w}$ | 2 |
| (d) w^2 | 1 |
| (e) $\overline{zw}$ | 2 |

Question 2 9 Marks

Given $z = 1 + i\sqrt{3}$ and $w = 1 + i$

- | | |
|---|---|
| (a) Find zw in $x + iy$ form. | 1 |
| (b) Convert z and w to modulus-argument form | 4 |
| (c) Hence find zw in modulus-argument form and use this to find the exact value of $\sin \frac{7\pi}{12}$. | 4 |

Question 3 5 Marks

- | | |
|---|---|
| (a) Convert $1 - i$ to mod/arg form. | 2 |
| (b) Hence find $(1 - i)^{11}$ in $x + iy$ form. | 3 |

Question 4 7 Marks

- (a) Find all real numbers x and y such that $\sqrt{-5+12i} = x+iy$ 5
- (b) Hence (or otherwise) solve for z : $z^2 + (2+i)z + (2-2i) = 0$. 2

Question 5 10 Marks

Sketch these regions on separate Argand diagrams:

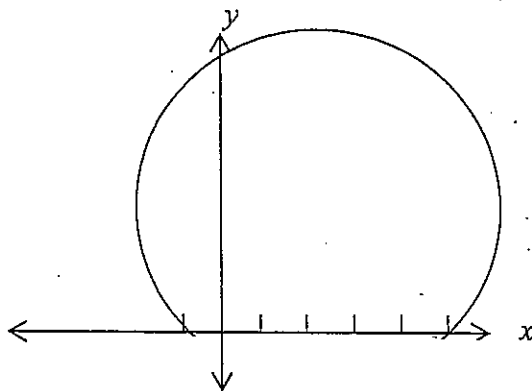
- (a) $|z-2| < 3$ 2
- (b) $\frac{-\pi}{3} < \text{Arg}(z+1) \leq \frac{\pi}{2}$ 3
- (c) $3 < |z| < 4$ and $\text{Arg}(z) > \frac{\pi}{2}$ 2
- (d) $|z-\bar{z}| < 3$ and $z\bar{z} \leq 5$ 3

Question 6 10 marks

(a) Find the Cartesian equations for the following loci:

- (i) $|z+2-5i| = 3$ 1
- (ii) $|z-1+2i| = |z+1+i|$ 2
- (iii) $\text{Re}\left(\frac{z-8i}{z+6}\right) = 0$ 3
- (b) The locus $\text{Arg}\left(\frac{z-5}{z+1}\right) = \frac{\pi}{3}$ represents part of a circle 4

(as shown). Find the centre and radius of the circle. Hence write the locus of z in Cartesian form.



Question 7 13 Marks

(a) z is an arbitrary complex number such that $|z| = 1$

and $0 < \text{Arg}(z) < \frac{\pi}{2}$. Copy the diagram and sketch on it

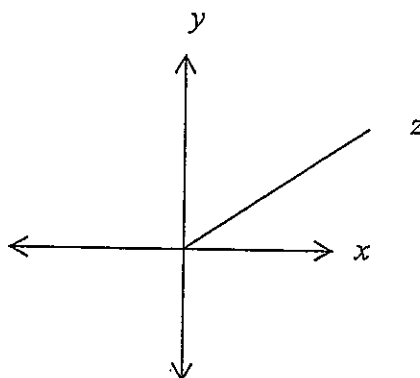
(labelling clearly)

(i) iz 1

(iii) $-z$ 1

(ii) $\bar{z}$ 1

(iv) $z+1$ 1



(b) For z in the diagram above (so $|z| = 1$) show that if $\text{Arg}(z) = \theta$ 2

then $\text{Arg}(z+1) = \frac{\theta}{2}$ (Hint: Use basic geometry!)

(c) z_1 is an arbitrary point on the complex plane.

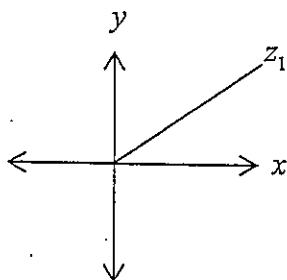
z_2 is a point on the complex plane such that $|z_1| = |z_2|$

and $\text{Arg}(z_2) = \frac{2\pi}{3} + \text{Arg}(z_1)$.

(i) Copy out the diagram below marking the points 3

$z_1, z_2, z_1 + z_2$ and $z_1 - z_2$.

(ii) Prove that $|z_1 + z_2| = |z_1|$. 4



Question 8 4 Marks

Prove DeMoivre's theorem i.e. $[(\cos \theta + i \sin \theta)]^n = (\cos n\theta + i \sin n\theta)$ for all positive integers n using the method of mathematical induction. In this proof you may NOT assume that $(\cos \alpha + i \sin \alpha) \times (\cos \beta + i \sin \beta) = [\cos(\alpha + \beta) + i \sin(\alpha + \beta)]$

Question 9 11 Marks

(a) If $z = \cos \theta + i \sin \theta$ prove

(i) $z^n + \frac{1}{z^n} = 2 \cos n\theta$ 1

(ii) $z^n - \frac{1}{z^n} = 2i \sin n\theta$ 1

(b) By letting $z = \cos \theta + i \sin \theta$ and expanding z^5 prove that

(i) $\sin 5\theta = 16 \sin^5 \theta - 20 \sin^3 \theta + 5 \sin \theta$ 3

(ii) $\cos 5\theta = 16 \cos^5 \theta - 20 \cos^3 \theta + 5 \cos \theta$ 1

(iii) Using your *workings* for parts (i) and (ii) find an expression for $\tan 5\theta$. 2

(c) Prove that $\sin^5 \theta = \frac{1}{16} [\sin 5\theta - 5 \sin 3\theta + 10 \sin \theta]$ 3

Question 10 22 Marks

(a)(i) Solve the equation $z^9 - 1 = 0$ over the complex field leaving your solutions in mod/arg form. 2

(ii) Find the area of the nonagon formed by joining the roots of $z^9 - 1 = 0$ on the complex plane. Answer correct to 2 decimal places. 1

(b) Let w be the non real root of $z^9 - 1 = 0$ with the smallest positive argument.

(i) Show that $w^2, w^3, w^4, w^5, w^6, w^7, w^8$ are the other non real roots of $z^9 - 1 = 0$. 2

(ii) Show that $w + w^2 + w^3 + w^4 + w^5 + w^6 + w^7 + w^8 = -1$. 2

(iii) Show that $(w + w^8)(w^2 + w^7)(w^4 + w^5) = -1$ 2

(iv) Hence show that $\cos \frac{\pi}{9} \cos \frac{2\pi}{9} \cos \frac{4\pi}{9} = \frac{1}{8}$ 4

PTD →

(c)(i) Resolve $z^5 - 1$ into real linear and quadratic factors. 2

(Leave your quadratic factors in terms of $\cos \theta$)

(ii) Hence show that $\cos \frac{2\pi}{5} + \cos \frac{4\pi}{5} = -\frac{1}{2}$ 3

$$\text{and } \cos \frac{2\pi}{5} \cos \frac{4\pi}{5} = -\frac{1}{4}$$

(iii) Hence find the exact value of $\cos \frac{2\pi}{5}$. 4

Solutions > Extension > p.1

Assessment Task 1 '08 Complex Numbers

Q. (1) (a) $z+w = -1+i$

(b) $\bar{z} = 1+4i$

(c) $\frac{z}{w} = \frac{1-4i}{-2+5i} \times \frac{(-2-5i)}{(-2-5i)}$

$= \frac{-22+3i}{29}$

(d) $w^2 = (-2+5i)^2$
 $= -21-20i$

(e) $\bar{z}w = (1-4i)(-2+5i)$
 $= 18+13i$
 $= 18-13i$

Q. (2) (a) $zw = (1+i\sqrt{3})(1+i)$
 $= (1-\sqrt{3}) + (1+\sqrt{3})i$

(b) $z = 2\text{cis}\frac{\pi}{3}, w = \sqrt{2}\text{cis}\frac{\pi}{4}$

(c) In mod/arg form
 $zw = 2\text{cis}\frac{\pi}{3} \times \sqrt{2}\text{cis}\frac{\pi}{4}$
 $= 2\sqrt{2}\text{cis}\frac{5\pi}{12}$
 $= 2\sqrt{2}\cos\frac{5\pi}{12} + 2i\sqrt{2}\sin\frac{5\pi}{12}$

Hence equating imaginary part with (a)

$2\sqrt{2}\sin\frac{5\pi}{12} = 1+\sqrt{3}$
 $\sin\frac{5\pi}{12} = \frac{1+\sqrt{3}}{2\sqrt{2}}$

(3) (a) $1-i = \sqrt{2}\text{cis}(-\frac{\pi}{4})$

(b) Hence $(1-i)^{11} = (\sqrt{2})^{11}\text{cis}(-\frac{11\pi}{4})$
 $= 32\sqrt{2}[\cos\frac{5\pi}{4} + i\sin\frac{5\pi}{4}]$
 $= 32\sqrt{2}[-\frac{1}{\sqrt{2}} - \frac{i}{\sqrt{2}}]$
 $= -32 - 32i$

(4) (a) $(x+iy)^2 = -5+12i$

① $x^2 + 2ixy - y^2 = -5+12i$

② Equating real parts: $x^2 - y^2 = -5$ (1)
 Equating imaginary parts: $2xy = 12$
 $\Rightarrow y = \frac{6}{x}$ (2)

Substituting (2) in (1):

$x^2 - (\frac{6}{x})^2 = -5$

$x^2 - \frac{36}{x^2} = -5$

× BS by x^2 & getting all to one side

$x^4 + 5x^2 - 36 = 0$

$(x^2+9)(x^2-4) = 0$ (5)

As x is real, $x = \pm 2$

As $y = \frac{6}{x}$, $y = \pm 3$

Square roots of $-5+12i$ are $2+3i$ and $-2-3i$

(b) Solving $z^2 + (2+i)z + (2-2i) = 0$

$z = \frac{-(2+i) \pm \sqrt{(2+i)^2 - 4 \times 1 \times (2-2i)}}{2 \times 1}$

$= \frac{-2-i \pm \sqrt{-5+12i}}{2}$

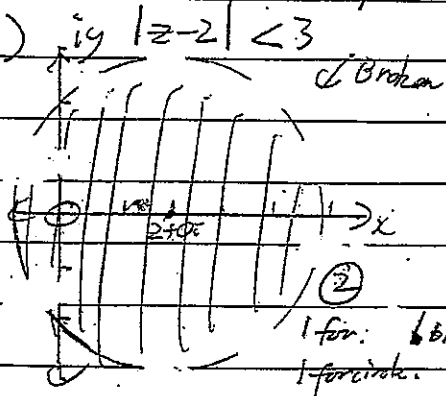
$= \frac{-2-i+2+3i}{2}$ or $= \frac{-2-i-2-3i}{2}$

Using (a)

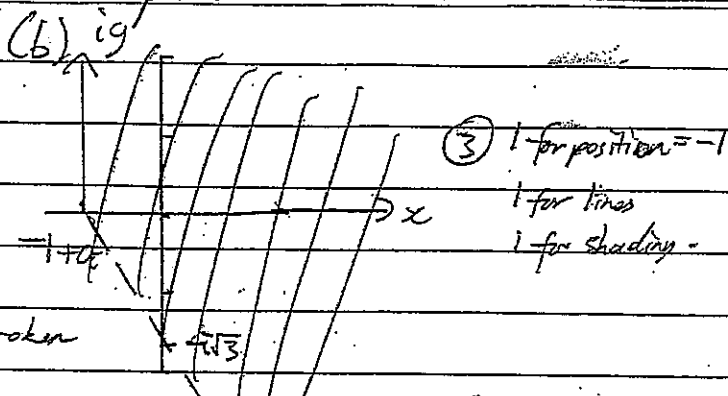
$= i$ or $= -2-2i$

$\therefore z = i$ or $z = -2-2i$

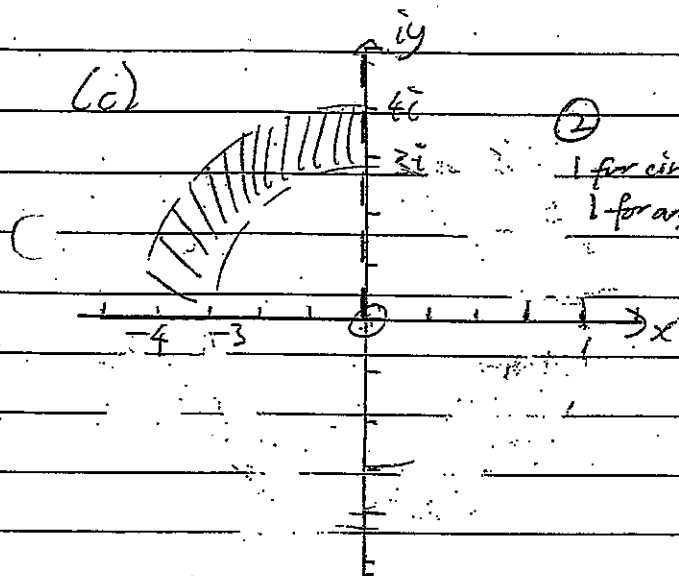
Q. (5) (a) $|z-2| < 3$



(b) $|z-2| < 3$

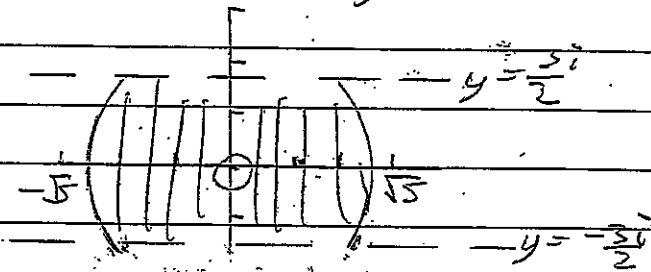


(c)



(d) $|z-\bar{z}| < 3$

$z-\bar{z} = (x+iy)-(x-iy) = 2iy$
 $|2iy| = |2y|$ So $|2y| < 3$
 $\Rightarrow -\frac{3}{2} < y < \frac{3}{2}$
 $\Rightarrow z^2 + y^2 \leq 5$



Q. (6) (a) (i) $(x+2)^2 + (y-5)^2 = 9$ ①

(ii) $\sqrt{(x-1)^2 + (y+2)^2} = \sqrt{(x+1)^2 + (y+1)^2}$

$x^2 - 2x + y^2 + 4y + 5 = x^2 + 2x + y^2 + 2y + 2$

$2y = 4x - 3$

$4x - 2y - 3 = 0$

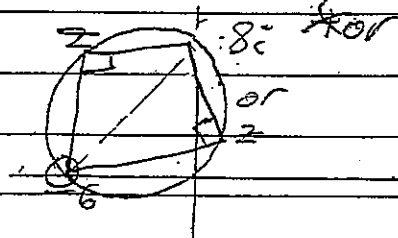
(iii) If $\text{Re}\left(\frac{z-8i}{z+6}\right) = 0$

then $\text{Arg}\left(\frac{z-8i}{z+6}\right) = \frac{\pi}{2}$ or $\frac{3\pi}{2}$

So $8i$ & 6 would be the endpoints of the diameter of a circle [in semicircle is a right angle].

(iii) [continued] So circle is

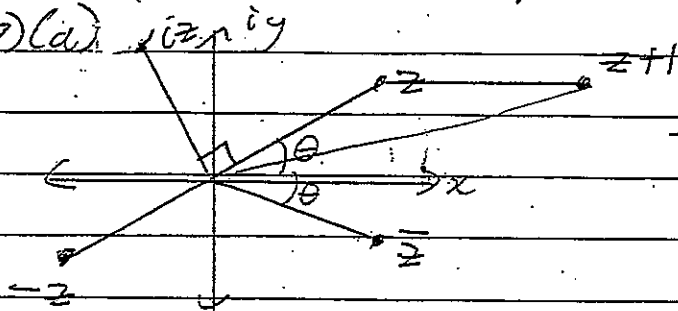
$(x+3)^2 + (y-4)^2 = 25$ NOT INCLUDED



[Note: $8i$ can be included.]

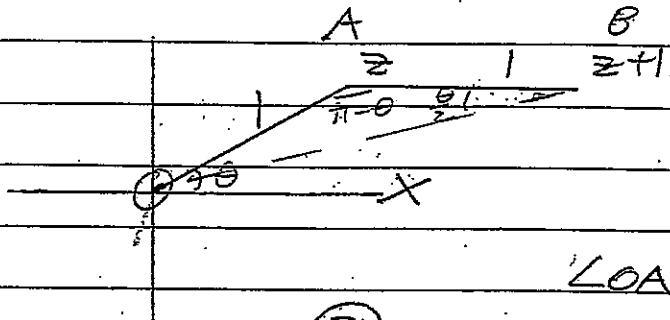
Ext. 2 Complex Solutions p. 4 '08

Q. (7)(a)



→ (4) lead.

(b)



Let z be at A

& $z+1$ be at B

Let $\text{Arg } z = \theta$.

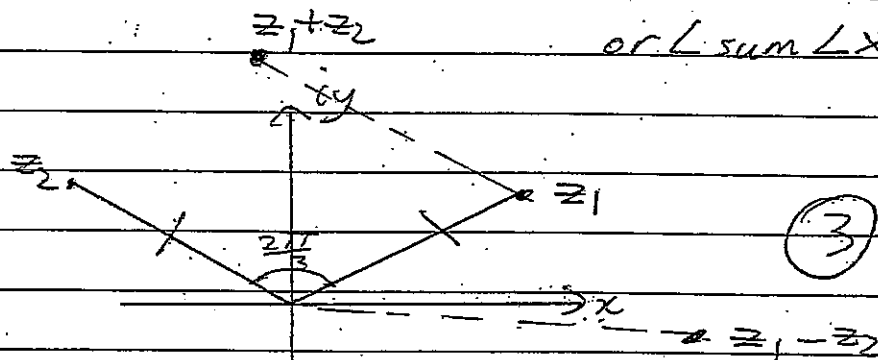
$\angle OAB = \pi - \theta$ [co-interior $\angle$'s, $OX \parallel AB$].

$\angle ABO = \angle AOB = \frac{\theta}{2}$ [$\angle$'s opposite = sides of isosceles $\triangle OAB$].

$\therefore \angle XO B = \frac{\theta}{2}$ [alternate $\angle$'s, $OX \parallel AB$ or $\angle$ sum $\angle XO A$].

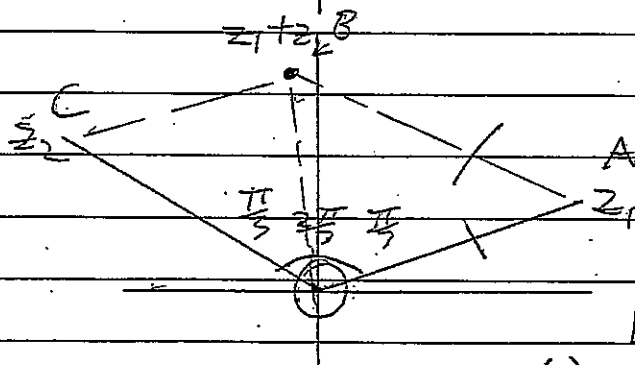
(c)

(i)



(3)

(ii)



Let $A = z_1, B = z_1 + z_2, C = z_2$

$OABC$ is a rhombus (all sides = 1).

OB bisects $\angle AOC$

[diagonals of rhombus bisect $\angle$'s they pass through].

$\therefore \angle AOB = \frac{\pi}{3}$.

(4)

$\therefore \triangle AOB$ is equilateral.

$\therefore |z_1 + z_2| = |z_1|$ [sides of equilateral $\triangle$].

Q.(8) Step 1: Show true for $n=1$:

(4)

$$(\cos \theta + i \sin \theta)^1 = \cos \theta + i \sin \theta$$

$\therefore$ True for $n=1$

Step 2: Assume true for $n=k$

$$\text{i.e. } (\cos \theta + i \sin \theta)^k = \cos k\theta + i \sin k\theta$$

Step 3: Prove true for $n=k+1$

$$\begin{aligned} \text{i.e. } (\cos \theta + i \sin \theta)^{k+1} &= \cos(k+1)\theta + i \sin(k+1)\theta \\ &= (\cos k\theta \cos \theta - \sin k\theta \sin \theta) + i(\sin k\theta \cos \theta + \cos k\theta \sin \theta) \end{aligned}$$

LHS :

$$\begin{aligned} &(\cos \theta + i \sin \theta)^{k+1} \\ &= (\cos \theta + i \sin \theta)^k \times (\cos \theta + i \sin \theta) \\ &= (\cos k\theta + i \sin k\theta) \times (\cos \theta + i \sin \theta) \\ &= \cos k\theta \cos \theta + i \cos k\theta \sin \theta + i \sin k\theta \cos \theta - \sin k\theta \sin \theta \\ &= (\cos k\theta \cos \theta - \sin k\theta \sin \theta) + i(\sin k\theta \cos \theta + \cos k\theta \sin \theta) \\ &= \text{RHS} \end{aligned}$$

$\therefore$ As it is true for $n=1$ & $n=k+1$ it must be true for all positive integers n by the principle of mathematical induction.

$$Q. (a)(i) z^n + \frac{1}{z^n}$$

$$= (\cos n\theta + i \sin n\theta) + (\cos n\theta - i \sin n\theta) \\ = 2 \cos n\theta. \quad (1)$$

$$(ii) z^n - \frac{1}{z^n}$$

$$= \cos n\theta + i \sin n\theta - (\cos n\theta - i \sin n\theta) \\ = 2i \sin n\theta \quad (1)$$

$$(b)(i) (\cos \theta + i \sin \theta)^5 = \cos 5\theta + i \sin 5\theta \\ \cos^5 \theta + 5i \cos^4 \theta \sin \theta - 10 \cos^3 \theta \sin^2 \theta - 10i \cos^2 \theta \sin^3 \theta + 5 \cos \theta \sin^4 \theta + i \sin^5 \theta$$

Hence equating imaginary parts: $\sin 5\theta = 5 \cos^4 \theta \sin \theta - 10 \cos^2 \theta \sin^3 \theta + \sin^5 \theta$ (2)

$$= 5(1 - \sin^2 \theta)^2 \sin \theta - 10(1 - \sin^2 \theta) \sin^3 \theta + \sin^5 \theta \\ = 5 \sin \theta (1 - 2 \sin^2 \theta + \sin^4 \theta) - 10(1 - \sin^2 \theta) \sin^3 \theta + \sin^5 \theta \\ = 16 \sin^5 \theta - 20 \sin^3 \theta + 5 \sin \theta \quad (3)$$

(ii) Equating real parts of (1):

$$\cos 5\theta = \cos^5 \theta - 10 \cos^3 \theta \sin^2 \theta + 5 \cos \theta \sin^4 \theta \quad (3) \\ = \cos^5 \theta - 10 \cos^3 \theta (1 - \cos^2 \theta) + 5 \cos \theta (1 - 2 \cos^2 \theta + \cos^4 \theta) \\ = 16 \cos^5 \theta - 20 \cos^3 \theta + 5 \cos \theta. \quad (1)$$

(iii) Using equations (2) & (3) from parts (i) & (ii) above:

$$\tan 5\theta = \frac{\sin 5\theta}{\cos 5\theta} = \frac{5 \cos^4 \theta \sin \theta - 10 \cos^2 \theta \sin^3 \theta + \sin^5 \theta}{\cos^5 \theta - 10 \cos^3 \theta \sin^2 \theta + 5 \cos \theta \sin^4 \theta}$$

Dividing numerator & denominator by $\cos^5 \theta$

$$= \frac{5 \frac{\sin \theta}{\cos \theta} - 10 \frac{\sin^3 \theta}{\cos^3 \theta} + \frac{\sin^5 \theta}{\cos^5 \theta}}{1 - 10 \frac{\sin^2 \theta}{\cos^2 \theta} + 5 \frac{\sin^4 \theta}{\cos^4 \theta}} \quad (2) \\ = \frac{5 \tan \theta - 10 \tan^3 \theta + \tan^5 \theta}{1 - 10 \tan^2 \theta + 5 \tan^4 \theta} \\ = \text{RHS.}$$

Ext. 2 Complex 28 p. 7 Solutions

Q. (9)(a) If $z = \cos \theta + i \sin \theta$

$$z - \frac{1}{z} = 2i \sin \theta$$

$$\left(z - \frac{1}{z}\right)^5 = 32i \sin^5 \theta \quad (i^5 = i) \quad \therefore 1$$

$$z^5 - 5z^3 + 10z - 10 + \frac{5}{z} - \frac{1}{z^5} = 32i \sin^5 \theta \quad (3)$$

$$\left(z^5 - \frac{1}{z^5}\right) - 5\left(z^3 - \frac{1}{z^3}\right) + 10\left(z - \frac{1}{z}\right) = 32i \sin^5 \theta \quad |$$

$$\text{As } z^n - \frac{1}{z^n} = 2i \sin n\theta$$

$$\left. \begin{aligned} 2i \sin 5\theta - 10i \sin 3\theta + 20i \sin \theta &= 32i \sin^5 \theta \\ \frac{1}{16} [\sin 5\theta - 5 \sin 3\theta + 10 \sin \theta] &= \sin^5 \theta \end{aligned} \right\}$$

Q. (10)(a)(i) $z^9 - 1 = 0$

$$z = 1, \text{cis } \frac{2\pi}{9}, \text{cis } \frac{4\pi}{9}, \text{cis } \frac{6\pi}{9}, \text{cis } \frac{8\pi}{9}, \text{cis } \frac{10\pi}{9}, \text{cis } \frac{12\pi}{9}, \text{cis } \frac{14\pi}{9}, \text{cis } \frac{16\pi}{9} \quad (2)$$

$\rightarrow 1$ for $\text{cis } \frac{2\pi}{9}$
 $\rightarrow 1$ for rest.

$$\begin{aligned} \text{(ii) Area} &= 9 \times \frac{1}{2} \times \sin \frac{2\pi}{9} \\ &= 2.89 \text{ square units (2DP)} \quad (1) \end{aligned}$$

$$(10) (b)(i) w = \text{cis} \left(\frac{2\pi}{9} \right) \Rightarrow w^2 = \text{cis} \left(\frac{4\pi}{9} \right), w^3 = \text{cis} \left(\frac{6\pi}{9} \right), w^4 = \text{cis} \left(\frac{8\pi}{9} \right)$$

$$\begin{aligned} w^5 &= \text{cis} \frac{10\pi}{9} & w^6 &= \text{cis} \frac{12\pi}{9} & w^7 &= \text{cis} \frac{14\pi}{9} & w^8 &= \text{cis} \frac{16\pi}{9} \\ &[\text{or } \text{cis} -\frac{8\pi}{9}] & & [= \text{cis} \frac{4\pi}{3}] & & [= \text{cis} -\frac{4\pi}{9}] & & [= \text{cis} -\frac{2\pi}{9}] \\ & & & \text{or } \text{cis} -\frac{2\pi}{3} & & & & \end{aligned}$$

$\therefore w^2, w^3, w^4, w^5, w^6, w^7, w^8$ are other non-real roots.

(ii) Roots of $z^9 - 1 = 0$ are $1, w, w^2, w^3, w^4, w^5, w^6, w^7, w^8$

$$\begin{aligned} \text{By sum of roots: } 1 + w + w^2 + w^3 + w^4 + w^5 + w^6 + w^7 + w^8 &= 0 \\ w + w^2 + w^3 + w^4 + w^5 + w^6 + w^7 + w^8 &= -1. \end{aligned}$$

Extr. 2 Complex '08 p. 8 Solutions:

Q. (10)(b)(iv) As $w = \text{cis } \frac{2\pi}{9}$

$$\begin{aligned} w + w^8 &= \left(\cos \frac{2\pi}{9} + i \sin \frac{2\pi}{9} + \cos \frac{16\pi}{9} + i \sin \frac{16\pi}{9} \right) \\ &= \left(\cos \frac{2\pi}{9} + i \sin \frac{2\pi}{9} + \cos \frac{2\pi}{9} - i \sin \frac{2\pi}{9} \right) \\ &= 2 \cos \frac{2\pi}{9} \end{aligned}$$

Similarly $w^2 + w^7 = 2 \cos \frac{4\pi}{9}$, $w^4 + w^5 = 2 \cos \frac{8\pi}{9} = -2 \cos \frac{\pi}{9}$

$\therefore$ As $(w + w^8)(w^2 + w^7)(w^4 + w^5) = -1$

$$2 \cos \frac{2\pi}{9} \times 2 \cos \frac{4\pi}{9} \times -2 \cos \frac{\pi}{9} = -1 \quad (4)$$

$$\cos \frac{2\pi}{9} \times \cos \frac{4\pi}{9} \times \cos \frac{\pi}{9} = \frac{1}{8}$$

(c) (i) $z^5 - 1 = (z - 1) \left(z - \text{cis } \frac{2\pi}{5} \right) \left(z - \text{cis } \frac{4\pi}{5} \right) \left(z - \text{cis } \frac{6\pi}{5} \right) \left(z - \text{cis } \frac{8\pi}{5} \right)$
 $= (z - 1) \left(z^2 - 2z \cos \frac{2\pi}{5} + 1 \right) \left(z^2 - 2z \cos \frac{4\pi}{5} + 1 \right)$

(ii) $\therefore$ As $z^5 - 1 = (z - 1)(z^4 + z^3 + z^2 + z + 1)$

$$\begin{aligned} z^4 + z^3 + z^2 + z + 1 &= \left(z^2 - 2z \cos \frac{2\pi}{5} + 1 \right) \left(z^2 - 2z \cos \frac{4\pi}{5} + 1 \right) \\ &= z^4 - 2z^3 \left(\cos \frac{2\pi}{5} + \cos \frac{4\pi}{5} \right) + \left(2 + 4 \cos \frac{2\pi}{5} \cos \frac{4\pi}{5} \right) z^2 \\ &\quad - 2z \left(\cos \frac{2\pi}{5} + \cos \frac{4\pi}{5} \right) + 1 \end{aligned} \quad (3)$$

Equating co-efficients of z^3 [or z]

$$-2z^3 \left(\cos \frac{2\pi}{5} + \cos \frac{4\pi}{5} \right) = 1$$

$$\cos \frac{2\pi}{5} + \cos \frac{4\pi}{5} = -\frac{1}{2}$$

$\rightarrow$ Notes This can be done using sum of roots of $z^5 - 1 = 0$ but a lot of explanation needed.

Equating co-efficients of z^2 :

$$2 + 4 \cos \frac{2\pi}{5} \cos \frac{4\pi}{5} = 1$$

$$\cos \frac{2\pi}{5} \cos \frac{4\pi}{5} = -\frac{1}{4}$$

(iii) Hence letting $x = \cos \frac{2\pi}{5}$, $y = \cos \frac{4\pi}{5}$

$$x + y = -\frac{1}{2} \quad (1) \quad \text{Sub. (2) in (1)}$$

$$xy = -\frac{1}{4} \quad (2) \quad x - \frac{1}{4x} = -\frac{1}{2}$$

$$y = -\frac{1}{4x} \quad 4x^2 + 2x - 1 = 0$$

$$x = \frac{-2 \pm \sqrt{2^2 - 4 \times 4 \times -1}}{2 \times 4}$$

$$= \frac{-2 \pm \sqrt{20}}{8}$$

$$= \frac{-2 \pm 2\sqrt{5}}{8}$$

$$x = \frac{-1 \pm \sqrt{5}}{4}$$

As $x = \cos \frac{2\pi}{5}$

& is in Q , $\cos$ is positive

$$\cos \frac{2\pi}{5} = \frac{-1 + \sqrt{5}}{4}$$

(4)